RYDBERG POSTER PRESENTATION

Speaking Script + Q&A Preparation

For the submitted poster: “Teaching Atoms to See Microwave Polarization”

Core presentation rule: separate what you measured from what you modelled or propose. Your strongest story is: measured linear-polarization discrimination → prove why linear-only analysis is mathematically incomplete → design a minimum four-state full-Stokes analyser → show the hardware and calibration pathway needed to realise it.

How to use this document

- Use the 6–8 minute script if you are given a normal poster walkthrough.
- Use the 90-second version if a judge says, “Give me the quick version.”
- For Q&A, lead with the one-sentence answer. Expand only if they keep engaging.
- Point to the relevant panel while answering. It makes technical answers easier to follow and buys you a second to think.

1. Main poster walkthrough (about 6–8 minutes)

OPENING — 20–30 seconds
“My thesis asks a simple question: can a Rydberg-atom receiver determine the full polarization state of an unknown microwave field? The receiver already shows clear sensitivity to linear polarization, but a full polarization measurement also needs the circular component. So the work on this poster moves from what we measured experimentally, to the mathematical limitation of linear measurements, and then to a four-state analyser architecture that can recover the full Stokes vector.”

Point to: title → Block 1 → sweep your hand across Blocks 3–8 to show the story arc.

Block 1 — Background and motivation (about 50 seconds)

“A Rydberg atom is an atom excited to a very high principal quantum number, so its outer electron is weakly bound and becomes extremely sensitive to RF electric fields. That sensitivity can be read optically through Rydberg EIT, which gives us a fundamentally different sensing mechanism from a conventional metal antenna.”

“The key question is not just whether the receiver sees RF power; it is whether it can recover polarization. Our present measurements distinguish linear orientations, but H, V, D and A measurements alone do not uniquely determine circular handedness. That leads to the central design question: what is the minimum analyser geometry that gives full-Stokes recovery?”

Emphasise the phrase “minimum analyser geometry” — it sets up Block 5.

Block 2 — How the atom converts RF into an optical measurement (about 55 seconds)

“The physical readout is a ladder-type Rydberg EIT system. A 780-nanometre probe and a 480-nanometre coupling laser prepare the optical transition. When the microwave field couples two Rydberg states, the EIT feature can split through the Autler–Townes effect.”

“In the resolved Autler–Townes regime, the splitting Δf_{AT} is proportional to the RF electric-field amplitude through the transition dipole moment. So the atom transfers information from the microwave field into an optical spectrum that we can detect with the photodetector.”

Point first to the experimental photograph, then the energy ladder, then the equation. Do not dwell on the equation unless asked.

Block 3 — Experimental proof of linear-polarization sensitivity (about 55 seconds)

“This is the experimental anchor of the poster. At 2 GHz, the 0°, 45° and 90° input states remain clearly ordered after the atomic readout. The 0° state gives the highest transmitted level at about −5.9 dB, the 90° state is strongly attenuated at about −15.6 dB, and the 45° point falls between them.”

“The solid curve is a Malus-law trend anchored to the measured 0° and 90° endpoints. So the important result here is not a fitted theoretical claim; it is that the atomic receiver preserves the expected ordering of linear polarization states.”

If time is short, quote only the 9.7 dB pass-to-reject separation and move on.

Block 4 — Why linear measurements are fundamentally incomplete (about 55 seconds)

“The next step is the key mathematical limitation. H, V, D and A analyser states all lie on the equator of the Poincaré sphere. They therefore span only the linear-polarization subspace. Two states can have the same S1 and S2 but opposite S3, which means opposite circular handedness, and a linear analyser cannot tell them apart.”

“The error map makes that limitation visual. If we force an arbitrary state onto S3 equals zero, the error is zero for linear states on the equator and grows toward 90 degrees at the two circular poles. So the missing circular component is not something that can be repaired by better numerical post-processing; the measurement basis itself is incomplete.”

This is one of your strongest conceptual panels. Say “measurement basis is incomplete,” not “the algorithm fails.”

Block 5 — Full-Stokes solution: tetrahedral analyser states (about 60 seconds)

“To recover the complete Stokes vector, we need analyser states that are non-coplanar on the Poincaré sphere. Four states are the minimum, and a regular tetrahedral arrangement gives a full-rank, well-conditioned analyser matrix.”

“The simulation on the right shows why that geometry matters under noise. Using 50,000 uniformly distributed fully polarized states, the tetrahedral basis gives a median Stokes-angle error of about 1.17° at 1% Gaussian readout noise. The linear rank-three basis remains near 29.77° even with zero added noise, because the missing S3 information never existed in the measurements.”

“The six-state canonical analyser can also perform well, but the tetrahedral basis achieves full-Stokes recovery with the minimum number of measurement states.”

If asked “why tetrahedral?”, answer: minimum state count + full rank + near-isotropic sensitivity.

Block 6 — What magnetic bias could add (about 50 seconds)

“We also investigated magnetic bias because Zeeman splitting can, in principle, separate sigma-minus, pi and sigma-plus pathways and add handedness sensitivity.”

“However, I am deliberately conservative about the experiment shown here. The magnetic-bias sweep spans 5.23 dB, while repeated zero-bias checks drift by 3.78 dB. That is enough to show sensitivity to operating conditions, but not enough to claim a robust magnetic-polarization signature.”

“So the right-hand panel is a reduced-order design model showing the behaviour we would want under stable bias; it is not a fit to the measured sweep.”

This wording is important. Do not say the magnetic-bias experiment proved handedness discrimination.

Block 7 — Hardware path: programmable metasurface (about 55 seconds)

“The practical route to the four-state analyser is an electronically reconfigurable metasurface placed in front of the same vapour-cell sensor. The goal is to switch between four calibrated analyser states without mechanically rotating the hardware.”

“The hardware success criteria are straightforward: all four states must transmit enough signal, the field should be sufficiently uniform over the sensing aperture, the calibrated analyser matrix must remain full rank and well conditioned, and the final reconstruction must work on held-out polarization states.”

If someone asks whether this exact metasurface has already completed full-Stokes validation, say no: this panel is the hardware pathway and design target.

Block 8 — From measured atomic splitting to the Stokes vector (about 60 seconds)

“The final panel closes the loop from the atomic measurement to the polarization estimate. For each analyser state, the measured Autler–Townes splitting is converted to a power-like observable q_k using its calibration constant.”

“After calibrating the four hardware states, those measurements form a response vector q. The experimentally calibrated analyser matrix is then inverted with a pseudoinverse to estimate the Stokes vector, which is finally normalised.”

“The validation experiment is to calibrate with known H, V, D, A, RHCP and LHCP references, reconstruct their Stokes vectors, and compare the recovered locations with their known positions on the Poincaré sphere. Agreement in both angular position and handedness is the criterion for successful full-Stokes operation.”

CLOSING — 20–30 seconds
“So the main conclusion is that the current Rydberg receiver already preserves measurable linear-polarization information, but linear-only measurements are mathematically incapable of recovering handedness. The thesis therefore identifies a minimum four-state tetrahedral analyser and a practical reconfigurable-metasurface route to full-Stokes sensing with a single vapour-cell receiver. The next decisive step is experimental calibration and validation of those four non-coplanar analyser states.”

2. 90-second version

“This poster is about turning a Rydberg-atom RF receiver into a full microwave polarimeter. The atom is read optically using Rydberg EIT: a microwave field couples Rydberg states, and in the resolved Autler–Townes regime the spectral splitting gives the RF field amplitude.

Experimentally, the receiver already preserves linear-polarization information. At 2 GHz, the 0°, 45° and 90° states remain ordered, with about 9.7 dB between the pass and reject cases.

But H, V, D and A measurements all lie on the equator of the Poincaré sphere, so they cannot recover S3 and cannot distinguish right- from left-hand circular polarization. That is a measurement-rank problem, not a post-processing problem.

The minimum full-Stokes solution is four non-coplanar analyser states. A regular tetrahedral basis is full rank and well conditioned, and the simulation shows low angular reconstruction error under noise. The proposed hardware route is an electronically switched four-state metasurface in front of one vapour cell. After calibration, the four atomic responses form a measurement vector that is inverted to recover the Stokes vector.

So the core contribution is the progression from experimentally verified linear sensitivity, to a proof of why linear analysis is incomplete, to a practical minimum-state architecture for full-Stokes Rydberg sensing.”

One-sentence version

“We experimentally show that a Rydberg receiver preserves linear-polarization information, prove why linear-only measurements cannot recover circular handedness, and design a minimum four-state analyser architecture for full-Stokes reconstruction.”

3. Q&A preparation

Answering strategy: give the first sentence, stop, and let the examiner decide whether they want the deeper explanation. Long answers should still end by reconnecting to the poster.

1. What is the main contribution of this work?

Short answer	The work connects three pieces: experimentally verified linear-polarization sensitivity in a Rydberg receiver, a rank-based proof that linear-only analyser states cannot recover circular handedness, and a minimum four-state architecture for full-Stokes reconstruction.
If they probe	The important distinction is that the linear discrimination is experimental, while the tetrahedral full-Stokes stage is the analyser design and validation pathway that the thesis develops from that result.

2. What exactly has been demonstrated experimentally?

Short answer	The poster experimentally demonstrates that the Rydberg readout preserves the ordering of prepared linear polarization states at 2 GHz.
If they probe	The 0° state is about −5.9 dB, the 90° state about −15.6 dB, and the 45° state lies between them, giving about 9.7 dB separation between the pass and reject cases. Full-Stokes handedness recovery is not claimed as already experimentally completed.

3. Why use Rydberg atoms instead of a conventional antenna?

Short answer	Rydberg atoms provide an optically read, atom-defined RF sensing mechanism with strong field sensitivity and very small sensing volumes.
If they probe	I would frame this as complementary rather than claiming that atoms universally replace classical antennas. The advantage is the different transduction mechanism and the possibility of minimally perturbing, calibration-driven sensing.

4. What is Rydberg EIT in simple terms?

Short answer	It is an optical transparency feature created when the probe and coupling lasers coherently connect the ground state to a Rydberg state.
If they probe	The microwave then perturbs or couples the Rydberg states, so microwave information appears in the optical spectrum measured by the photodetector.

5. What is Autler–Townes splitting and why does it matter here?

Short answer	When the RF coupling is strong enough to be spectrally resolved, it splits the EIT feature into two components.
If they probe	The splitting Δf_{AT} is proportional to the RF Rabi frequency, so with the relevant transition dipole moment it can be converted into RF electric-field amplitude.

6. How does polarization enter the atomic measurement?

Short answer	The RF coupling strength depends on the field component that drives the allowed atomic transition for the chosen quantization and analyser state.
If they probe	In practice, the analyser optics/metasurface and atomic selection rules determine how each input polarization maps into a measurable atomic response.

7. Why can H/V/D/A not recover full polarization?

Short answer	Because H, V, D and A all lie on the S3 = 0 equator of the Poincaré sphere.
If they probe	Their analyser matrix has rank 3 for the full Stokes problem, so S3 is missing. States with identical S1 and S2 but opposite S3 produce indistinguishable linear-only measurements.

8. Why is this not fixable with a better machine-learning or inversion algorithm?

Short answer	Because the missing circular component was never measured.
If they probe	No numerical method can uniquely reconstruct information that is absent from the measurement basis. The solution is to add non-coplanar analyser states, not merely a more sophisticated estimator.

9. What does S3 physically represent?

Short answer	S3 represents the circular component of the polarization state and carries handedness information.
If they probe	Its sign distinguishes the two circular senses under the adopted Stokes convention.

10. Why are four analyser states the minimum?

Short answer	A full Stokes vector has four components, so the analyser must provide four independent measurements.
If they probe	Geometrically, the four analyser states must be non-coplanar on the Poincaré sphere so that the analyser matrix is full rank.

11. Why choose a tetrahedral arrangement?

Short answer	A regular tetrahedron gives four non-coplanar states with good symmetry and conditioning.
If they probe	That distributes sensitivity more evenly over the Poincaré sphere and avoids making one polarization direction much less observable than another.

12. What does “well conditioned” mean in this context?

Short answer	It means small measurement noise produces relatively small errors in the reconstructed Stokes vector.
If they probe	A poorly conditioned analyser matrix amplifies noise strongly during inversion; a well-conditioned basis is much more robust.

13. Why not just use six canonical states H/V/D/A/RHCP/LHCP?

Short answer	You can; six canonical states are redundant and can also be robust.
If they probe	The tetrahedral basis is attractive because it reaches full rank with only four states, so it is the minimum-state solution and reduces switching and calibration burden.

14. Where does the 1.17° result come from?

Short answer	It comes from the Monte-Carlo robustness study shown in Block 5.
If they probe	The simulation used 50,000 uniformly distributed fully polarized states. At 1% Gaussian readout noise, the tetrahedral basis gives a median Stokes-angle error of about 1.17°.

15. Why does the linear basis show about 29.77° error even with zero added noise?

Short answer	Because the error is structural rather than noise-driven.
If they probe	The rank-three linear basis lacks S3, so some fully polarized states are inherently projected onto an incomplete measurement subspace.

16. Did the magnetic-bias experiment work?

Short answer	It showed sensitivity to operating conditions, but it did not provide strong enough evidence for a robust magnetic-polarization signature.
If they probe	The sweep spans 5.23 dB while repeated zero-bias checks drift by 3.78 dB, so the conservative conclusion is that the effect is not cleanly separable from drift in the present dataset.

17. Then why include the magnetic-bias panel?

Short answer	Because magnetic bias remains physically relevant as a possible auxiliary handedness-sensitive channel.
If they probe	The reduced-order model shows how stable Zeeman splitting could separate σ−, π and σ+ pathways, while the experimental panel transparently shows why the current dataset is insufficient to claim that result.

18. Is the right-hand magnetic-bias model a fit to the data?

Short answer	No.
If they probe	It is explicitly a reduced-order design map showing the expected qualitative Zeeman pathway separation under stable bias, not a fitted explanation of the experimental sweep.

19. What is the role of the metasurface?

Short answer	It is the proposed hardware analyser that generates four calibrated polarization-sensitive measurement states in front of the same vapour cell.
If they probe	The goal is electronic switching rather than mechanical rotation, so a single atomic receiver can acquire the independent measurements required for full-Stokes inversion.

20. Has the four-state metasurface already demonstrated full-Stokes recovery?

Short answer	Not in the result shown on this poster.
If they probe	Block 7 defines the hardware path and success criteria; Block 8 defines the calibration and validation experiment needed to close the loop.

21. What are the hardware success criteria?

Short answer	All four states need measurable transmission, adequate field uniformity over the sensing aperture, a full-rank and well-conditioned calibrated analyser matrix, and low reconstruction error on held-out states.
If they probe	Those criteria matter more than whether a single simulated S-parameter looks impressive.

22. How do you obtain the analyser matrix experimentally?

Short answer	Calibrate each analyser state using known reference polarizations and record the corresponding atomic responses.
If they probe	Those calibrated responses populate the analyser matrix. The matrix used in the final reconstruction should therefore be measured, not assumed from the ideal geometry.

23. What is q_k in Block 8?

Short answer	q_k is a power-like observable derived from the measured atomic response for analyser state k.
If they probe	On the poster it is written as the squared, calibrated Autler–Townes splitting, q_k = (Δf_AT,k / C_k)^2.

24. Why use the pseudoinverse?

Short answer	The pseudoinverse provides the least-squares linear reconstruction from the calibrated measurement vector to the Stokes estimate.
If they probe	It also handles the general case cleanly if the calibrated analyser matrix is not exactly ideal.

25. How would you validate the complete system?

Short answer	Calibrate the four hardware states, then test known H, V, D, A, RHCP and LHCP inputs that were not used as arbitrary unknowns during reconstruction.
If they probe	Success means that the reconstructed points land at the correct angular positions on the Poincaré sphere and that the sign of the circular component gives the correct handedness.

26. What is the biggest limitation of the current work?

Short answer	The strongest limitation is that the full four-state hardware validation has not yet been experimentally completed.
If they probe	The poster therefore separates the experimentally demonstrated linear result from the full-Stokes analyser design, simulation and validation pathway.

27. What would you do next if you had more lab time?

Short answer	I would stabilise the optical/RF operating point, fabricate or finalise the four-state reconfigurable analyser, calibrate all four states, and run blind reconstruction tests on known linear and circular references.
If they probe	I would also repeat the magnetic-bias study with improved drift control before making any claim about Zeeman-assisted handedness sensitivity.

28. Why use rubidium?

Short answer	Rubidium provides accessible optical transitions for laser excitation to Rydberg states and is widely used in vapour-cell Rydberg electrometry.
If they probe	For this project, it also matches the available probe/coupling laser architecture shown in the experiment.

29. Does your method measure phase directly?

Short answer	Not in the way a conventional coherent RF receiver directly measures I and Q.
If they probe	Instead, the polarization state is inferred from a calibrated set of intensity- or splitting-derived analyser responses. The independent analyser states encode the information needed to reconstruct the Stokes vector.

30. Is this a replacement for conventional RF receivers?

Short answer	I would describe it as a complementary sensing architecture rather than a universal replacement.
If they probe	Its value is in atom-based, optically read RF sensing and in operating regimes where small sensing volume, field perturbation, or a fundamentally different measurement mechanism is useful.

4. Hard questions and safe wording

“So have you actually demonstrated full-Stokes sensing?”

Say: “Not yet as a complete four-state hardware demonstration. The experimentally demonstrated result is linear-polarization discrimination. The full-Stokes contribution here is the rank analysis, optimal four-state analyser design, noise study, hardware path and calibration procedure.”

“Is the magnetic-bias result significant?”

Say: “It is suggestive of operating-condition sensitivity, but the drift is too large relative to the sweep to make a robust polarization claim. That is why I present the Zeeman panel as a design model rather than as a fit.”

“Why should I believe the tetrahedral basis if it is simulated?”

Say: “The full-rank and conditioning argument is geometric and linear-algebraic; the Monte-Carlo study then quantifies noise sensitivity. The remaining experimental question is whether the hardware can realise four sufficiently distinct calibrated states.”

“Your linear data only have three polarization angles. Is that enough?”

Say: “It is enough for the limited claim I make here: the atomic readout preserves the expected ordering of the prepared linear states and produces a strong pass-to-reject contrast. I do not claim that those three points alone constitute full polarimetric calibration.”

“Could the measured changes just be ordinary transmission through the analyser rather than the atom?”

Say: “The claim is that the prepared polarization dependence survives through the complete atomic readout chain. To isolate every contribution separately would require additional controlled measurements, but this result establishes that the receiver output retains polarization-dependent information.”

“Why not directly measure RHCP and LHCP now?”

Say: “Because the current linear analyser basis does not contain an independent S3-sensitive state. That is precisely the reason for the four-state non-coplanar analyser and the reconfigurable metasurface pathway.”

Phrases that keep your claims precise

	Recommended wording
Use	“demonstrates linear-polarization sensitivity”
Use	“shows the limitation of a linear-only analyser basis”
Use	“the tetrahedral solution is the minimum full-rank design”
Use	“proposed hardware path” / “validation pathway”
Use	“the magnetic-bias dataset is insufficient for a robust polarization claim”
Avoid	“we have already demonstrated full-Stokes experimentally”
Avoid	“magnetic bias proved handedness detection”
Avoid	“Rydberg atoms replace conventional antennas”

5. One-page mental cheat sheet

Remember	What to say
Problem	Linear discrimination works, but full polarization needs S3 / handedness.
Experiment	2 GHz linear states: 0° ≈ −5.9 dB, 45° ≈ −7.2 dB, 90° ≈ −15.6 dB; pass–reject ≈ 9.7 dB.
Physics	780 nm probe + 480 nm coupling → Rydberg EIT; RF coupling can produce Autler–Townes splitting.
Limitation	H/V/D/A are coplanar on the Poincaré equator → analyser rank 3 → S3 not observable.
Solution	Four non-coplanar analyser states; regular tetrahedron = minimum full-rank, well-conditioned basis.
Noise result	50,000 states; at 1% Gaussian readout noise, tetrahedral median angular error ≈ 1.17°.
Magnetic bias	Sweep span 5.23 dB vs zero-bias drift 3.78 dB → do not claim robust magnetic-polarization signature.
Hardware	Electronically switched four-state metasurface in front of one vapour cell.
Reconstruction	Calibrate four states → build q → apply calibrated analyser-matrix pseudoinverse → normalise Stokes vector.
Validation	Test H/V/D/A/RHCP/LHCP; success = correct angular position + correct handedness.

Final reminder: if you are challenged on what is “measured” versus “designed,” that is not a weakness if you answer clearly. The poster’s logic is strongest when you explicitly distinguish the measured linear result, the mathematical full-Stokes requirement, and the hardware experiment still needed to validate the complete analyser.